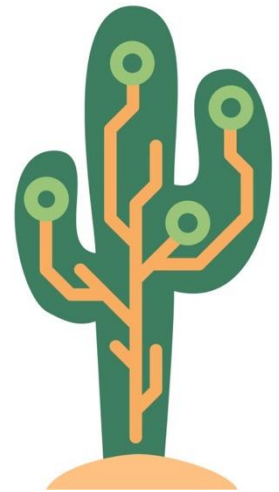


Advanced Hardware and Systems for Machine Learning

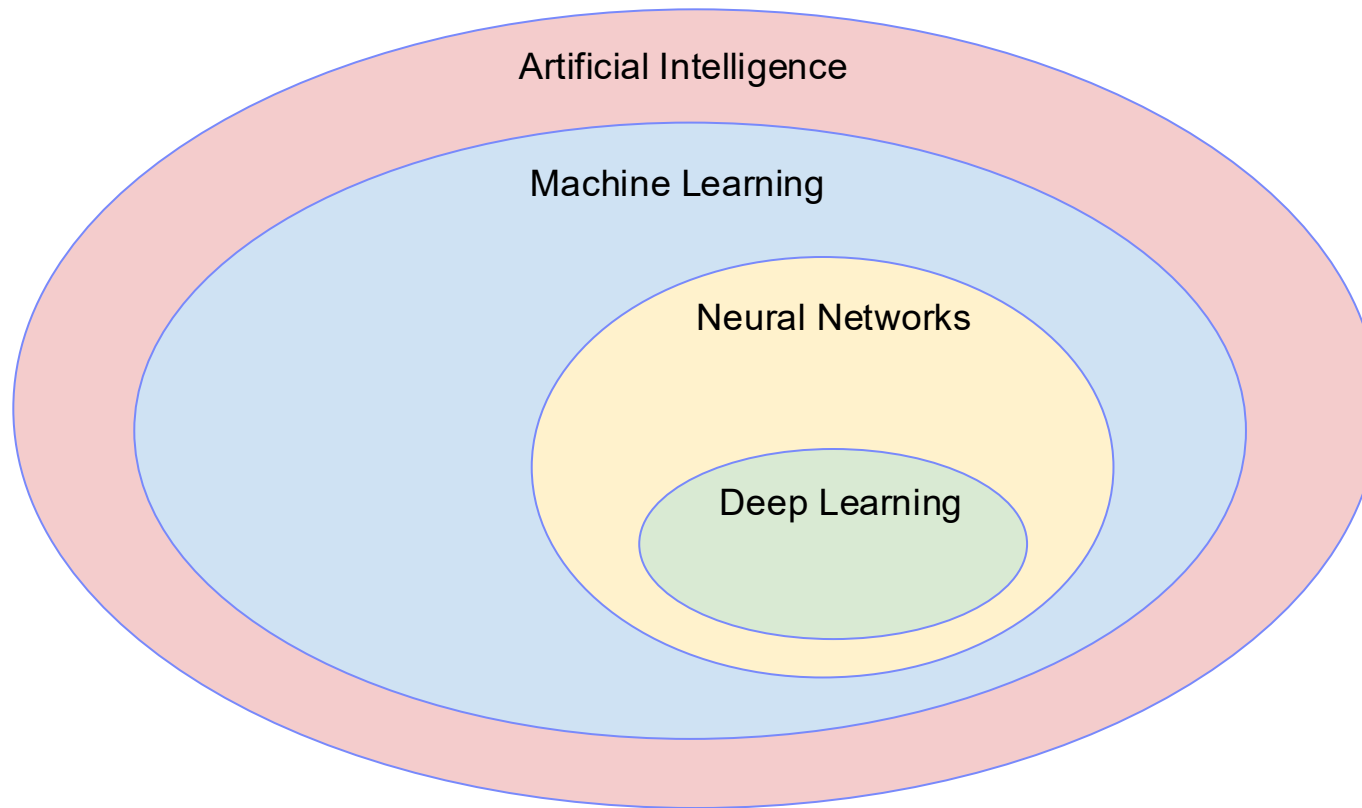
Lecture 2, 3: Machine Learning Basics



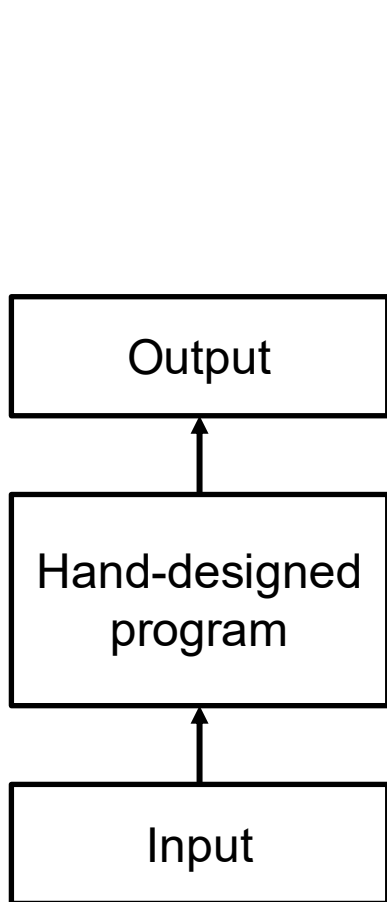
Jeff (Jun) Zhang
Assistant Professor
School of ECEE, ASU
jeffzhang@asu.edu



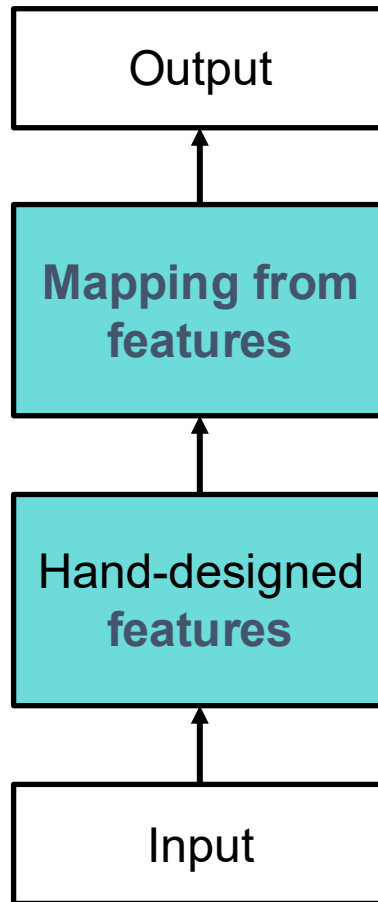
Deep Learning



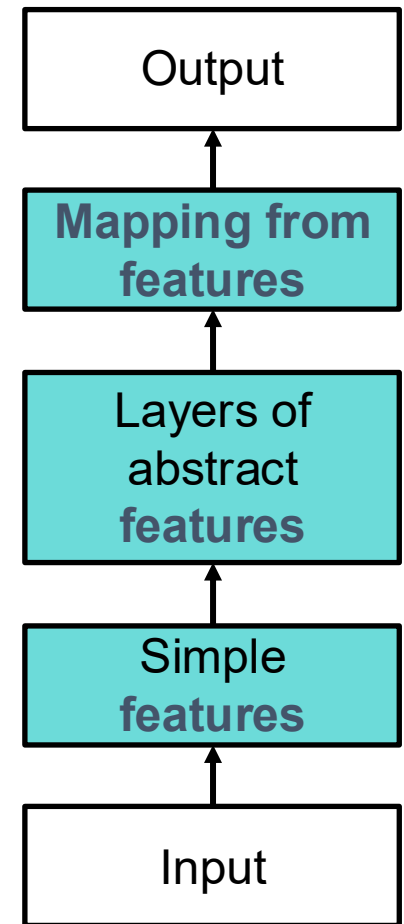
Different AI Algorithms



**Rule-based
system**

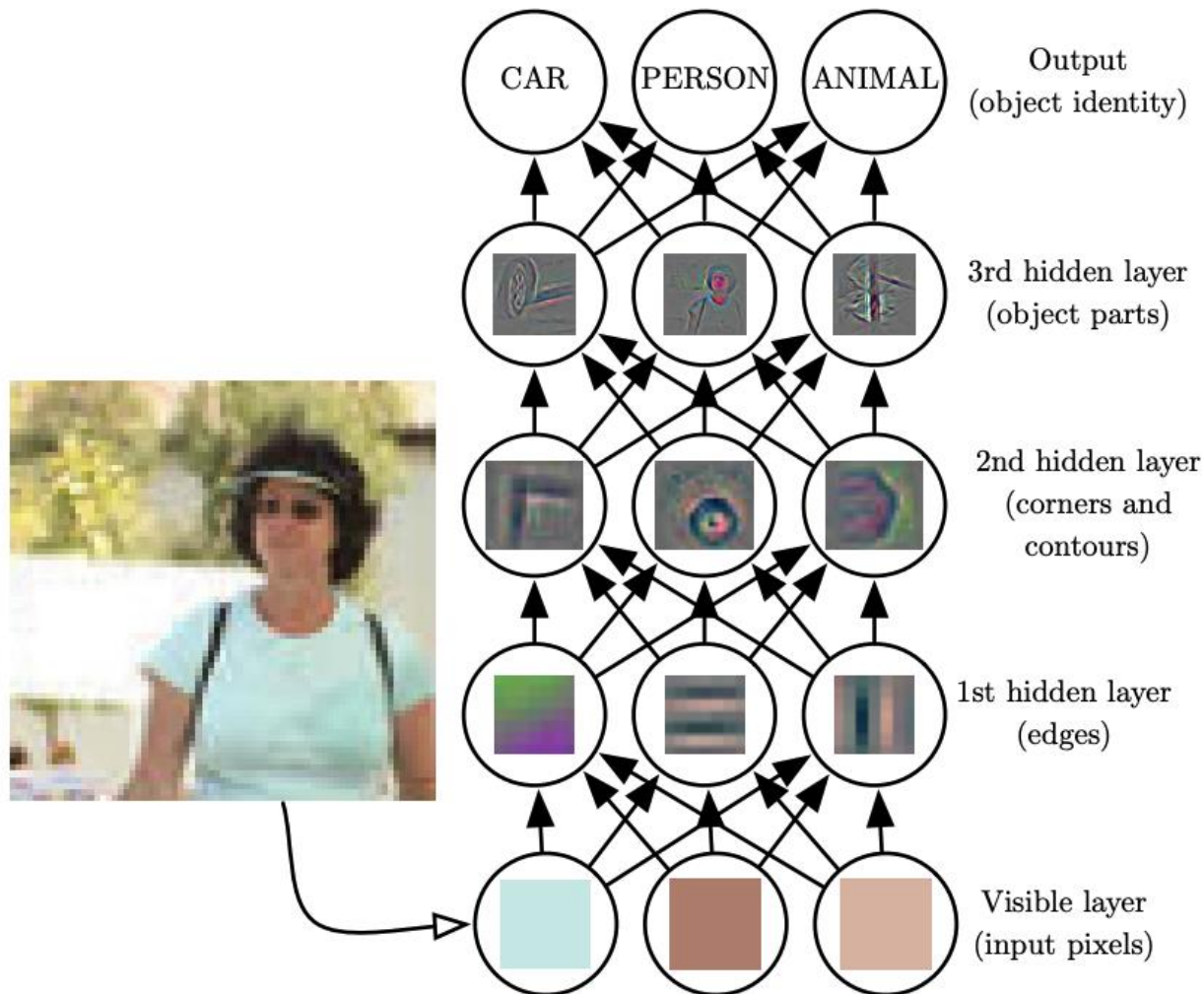


**Classic
machine
learning**



**Deep
learning**

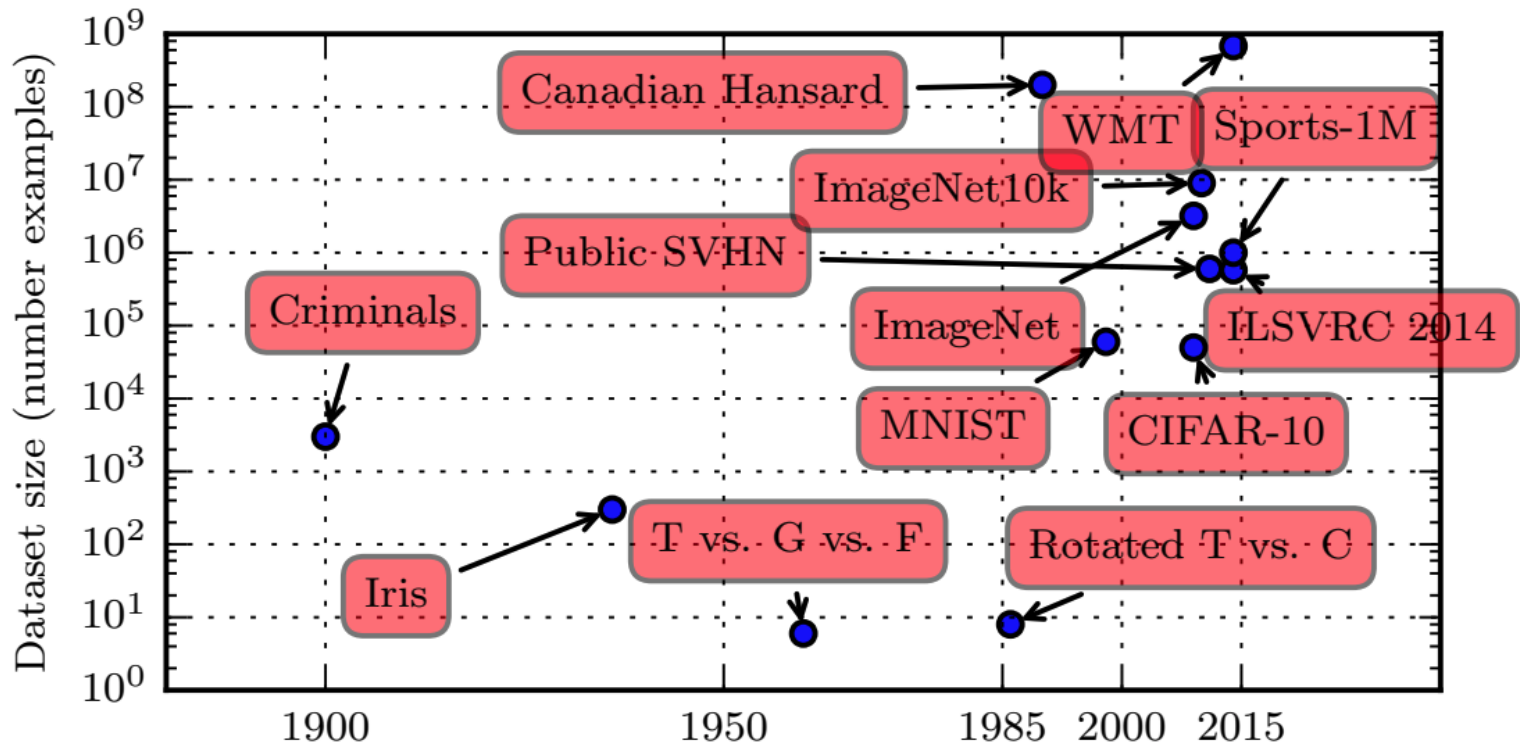
Deep Learning Example



Deep Learning, Goodfellow et. al.

Dataset Sizes

- The size of datasets has expanded remarkably over time.
- The age of “Big Data” has made machine learning easier.



Deep Learning, Goodfellow et. al.

A Toy Dataset: CIFAR 10

- **60,000 32x32 images**
 - 50,000 training
 - 10,000 test
- **10 classes**
 - 6,000 images/class

airplane

automobile

bird

cat

deer

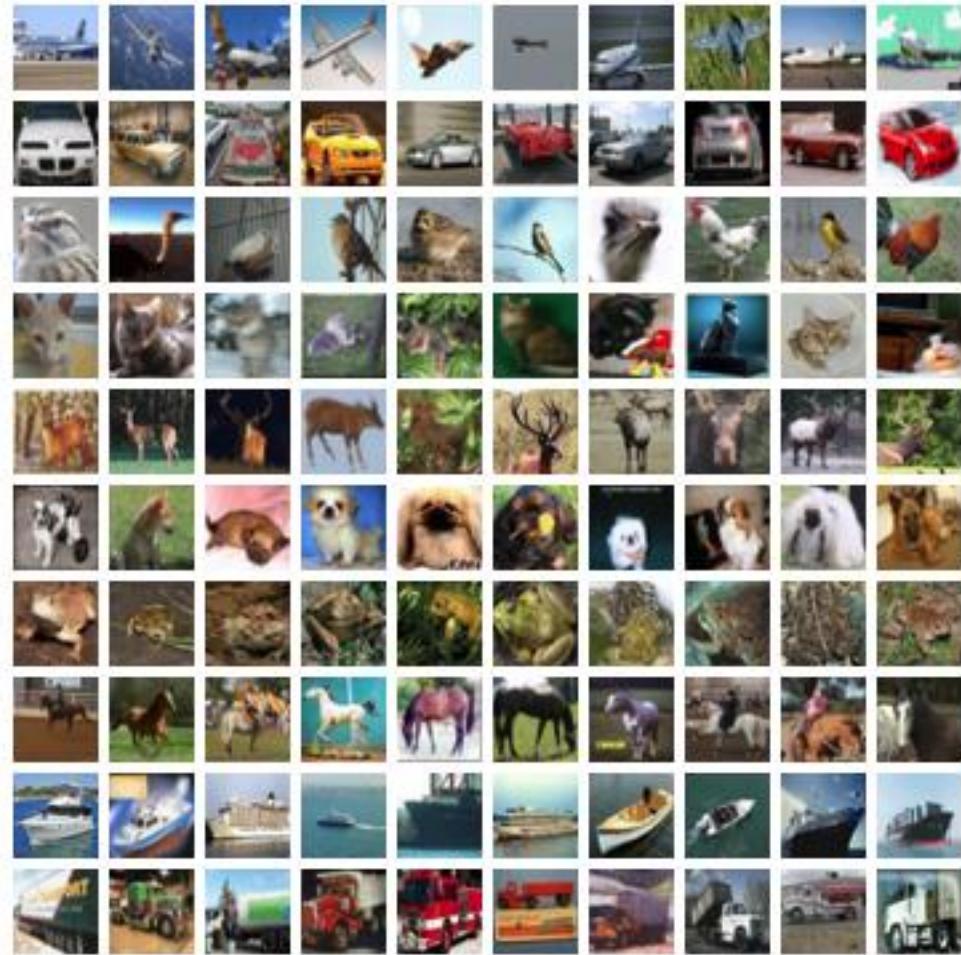
dog

frog

horse

ship

truck



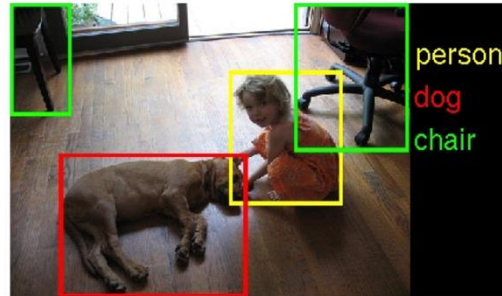
Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton

Dataset: ImageNet



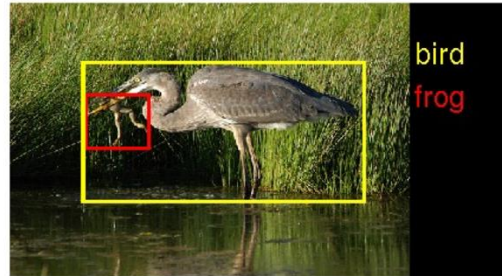
Image Classification Task

1.2M training images
1000 object categories



Object Detection Task

456k training images
200 object categories

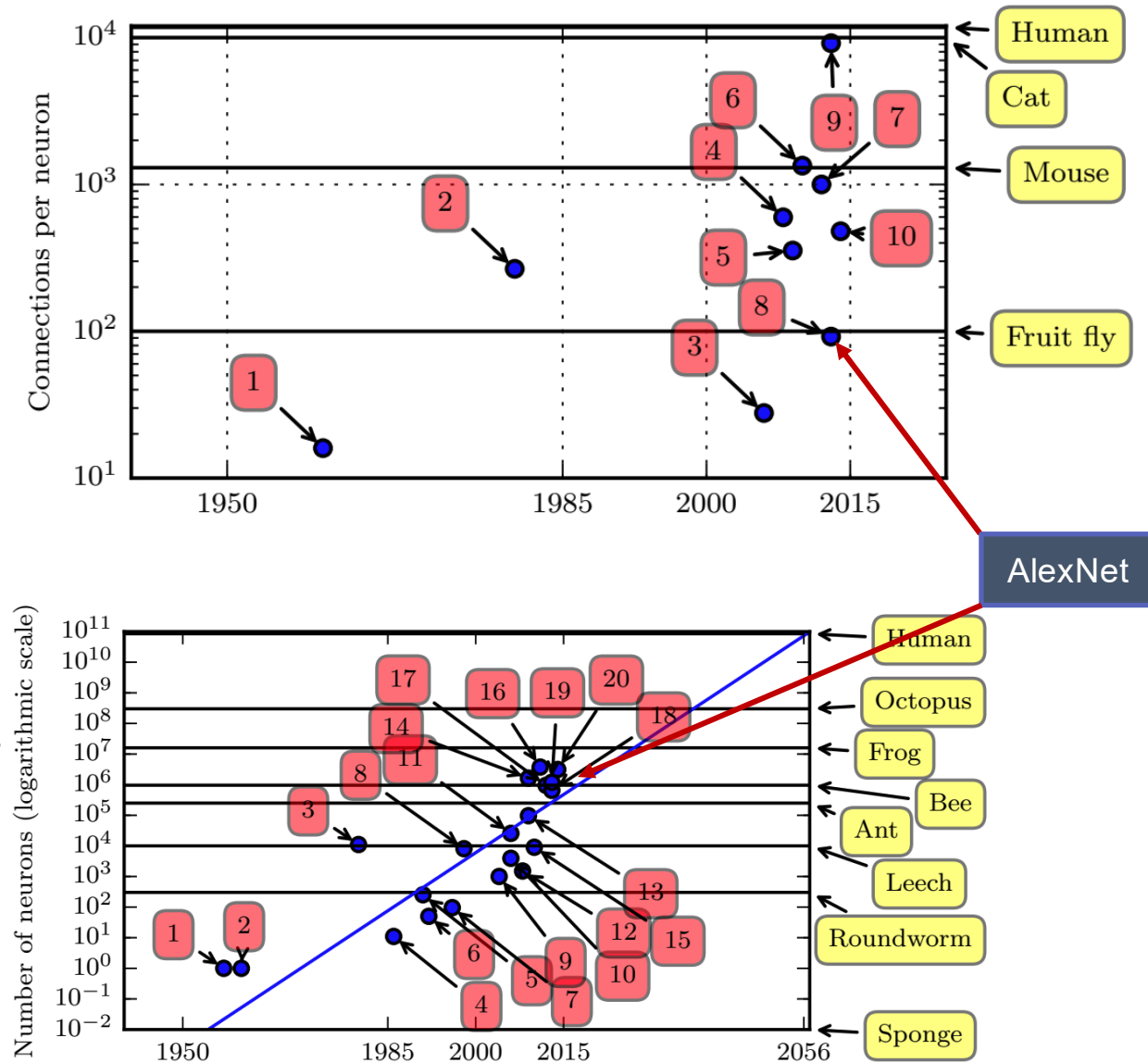


Model Sizes

■ Model size:

- # of connections / neuron
- # of neurons

- Largely due to the availability of faster hardware (CPUs and GPUs)
- Expect to continue



Deep Learning, Goodfellow et. al.

What is Machine Learning?

- Ability for machines to learn without being *explicitly* programmed

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E ." --- Mitchell, T. (1997). Machine Learning. McGraw Hill. p. 2.

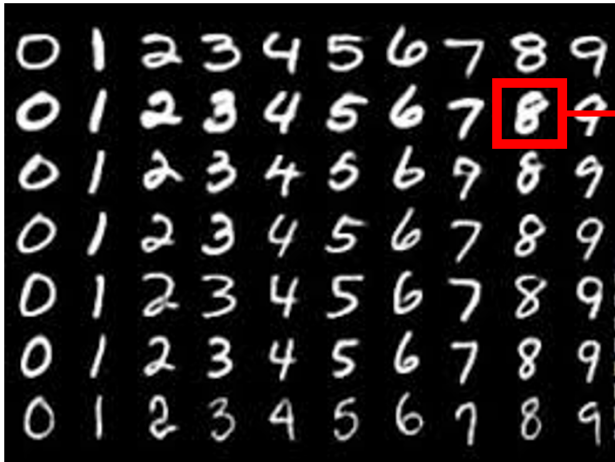
- Why not use user knowledge, experience or expertise?
 - Are humans always able to explain their expertise?
 - Can machines *outperform* humans?
- What kinds of experiences (E), tasks (T) and performance measures (P)?

Example: MNIST Digit Recognition

Task (T):

- Given gray-scale images $x \in [0,255]^{28 \times 28}$ and $y \in [0,9]$ find a function

$$f: x \rightarrow y$$



<https://www.npmjs.com/package/mnist>

Experience (E):

- A "training dataset" a set of correctly labeled images

Performance (P):

- Accuracy on a "test dataset"

"Supervised Learning (Classification)"

Example: Spam Classification

Task (T):

- Emails $x \in$ all possible emails and $y \in \{spam, non_spam\}$ find

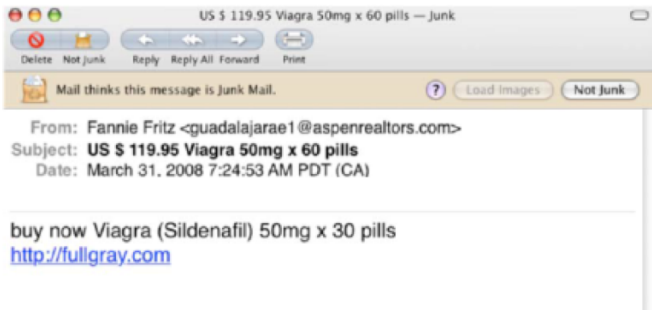
$$f: x \rightarrow y$$

Experience (E):

- A “**training dataset**” emails marked as “spam” or “non_spam”

Performance (P):

- Spam detection **accuracy**



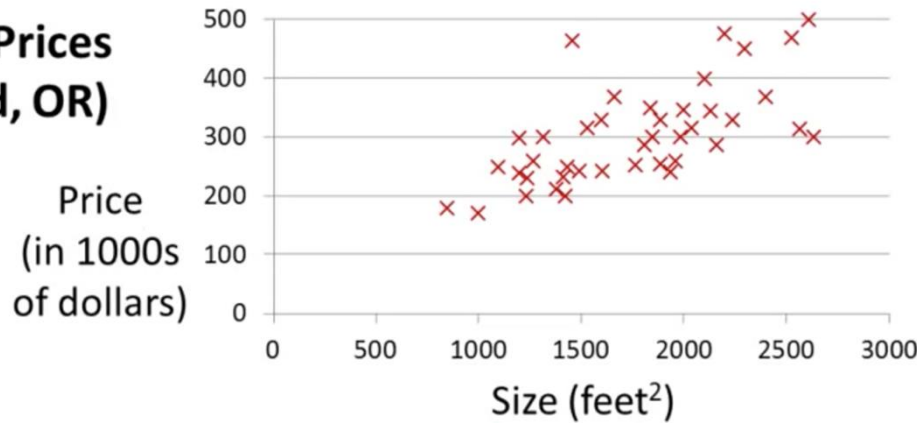
“Supervised Learning (Classification)”

Building a ML Algorithm

- **Nearly all ML algorithms can be described as particular instances of a simple recipe:**
 - A dataset -> Experience (E)
 - A cost (loss) function -> Performance Measure (P)
 - A model + An optimization method -> Task (T)
- **Use this recipe to see the different algorithms:**
 - As part of a taxonomy of methods for doing related tasks that work for similar reasons
 - Rather than as a long list of algorithms that each have separate justifications

Example: Linear Regression

Housing Prices (Portland, OR)



Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

■ Dataset:

- (x, y) where x is size and y is price
- m training examples

■ Cost function:

- Mean Squared error
- $MSE = \frac{1}{m} \sum_{i=0}^m (h(x_i) - y_i)^2$

■ Model:

$$h = w_0 + w_1 * x$$

■ Optimization method:

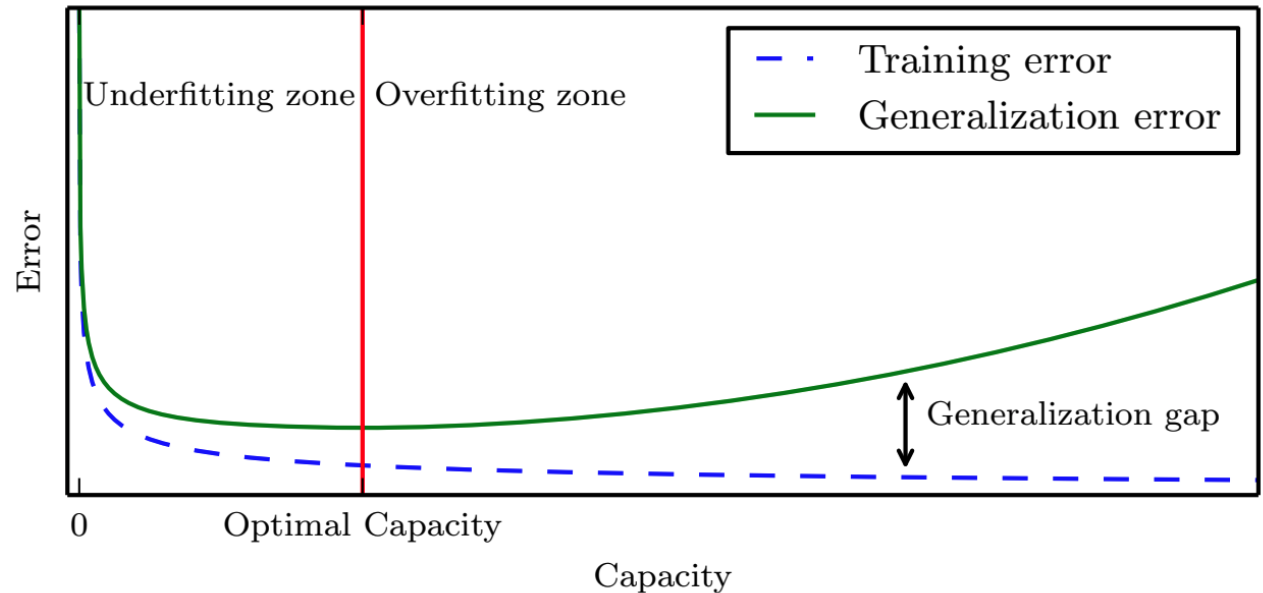
- Solve for where its gradient is 0.
- Gradient descent

Machine Learning, Andrew Ng

What is Dataset?

- ML tasks are usually described in terms of how the ML system should process an example.
- A dataset is a collection of many examples.
- ML algorithms can be broadly categorized as ***supervised*** and ***unsupervised*** by what kind of dataset they process.
 - Supervised: each example of the dataset is associated with a label or target
 - E.g., Classification, regression
 - Unsupervised: experience dataset without labels
 - E.g., Clustering
 - Reinforcement: Not a fixed dataset, interact with an environment

Dataset



- **Generalization:**

- Algorithm performs well on new, previously unseen inputs (not just those on which the model was trained).

- **Partition the dataset into:**

- Training set: where the model was trained
- Test set: where the trained model was tested

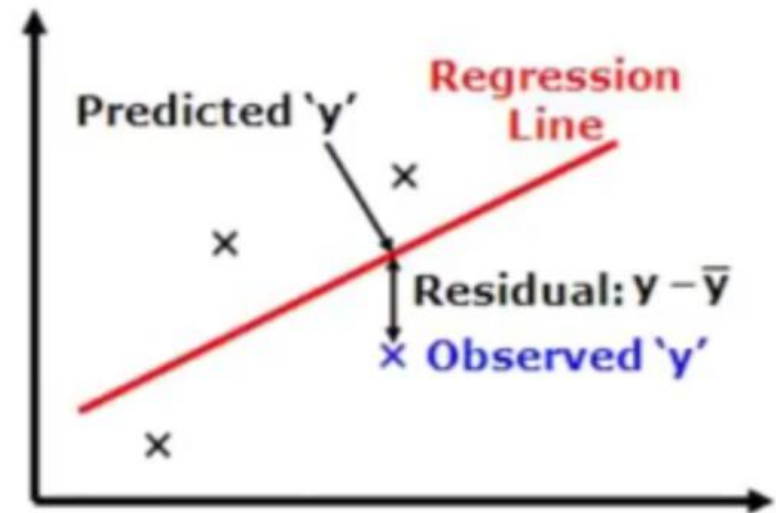
- **Overfitting:**

- Large gap between training and test errors

Cost function

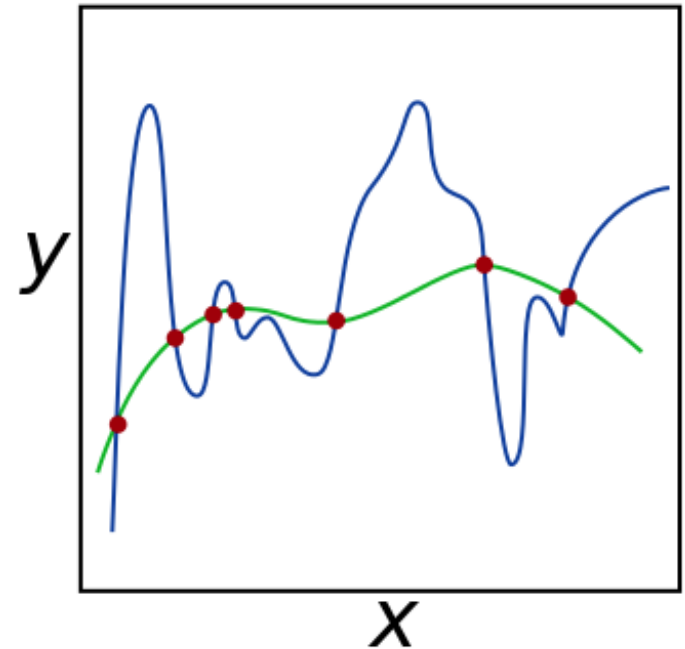
- Measures the performance of a ML model for given data
- Quantifies the error between predicted and expected values in the *training* set.

$$- CF = MSE_{train} = \frac{1}{m} \sum_{i=0}^m (\hat{y}_i - y_i)^2$$



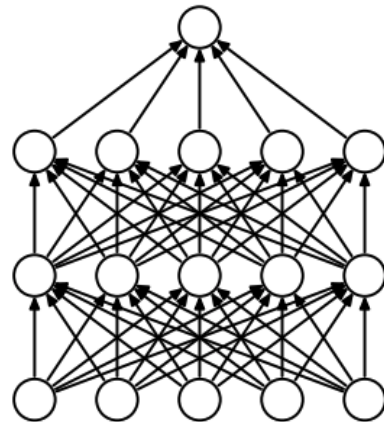
Cost function

- **Regularization: modifications to reduce the generalization error but not the training errors**
 - Weight decay (L2/L1 regularization)
 - Expressing preferences of smaller weights
 - $CF = MSE_{train} + \lambda \sum_i w_i^2$ (L2)
 - $CF = MSE_{train} + \lambda \sum_i |w_i|$ (L1)

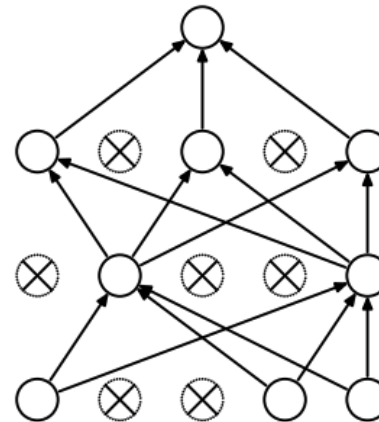


Dropout

- **Regularization: modifications to reduce the generalization error but not the training errors**
 - Dropout
 - Temporally remove nodes from network
 - Train a large ensemble of models that share parameters



(a) Standard Neural Net

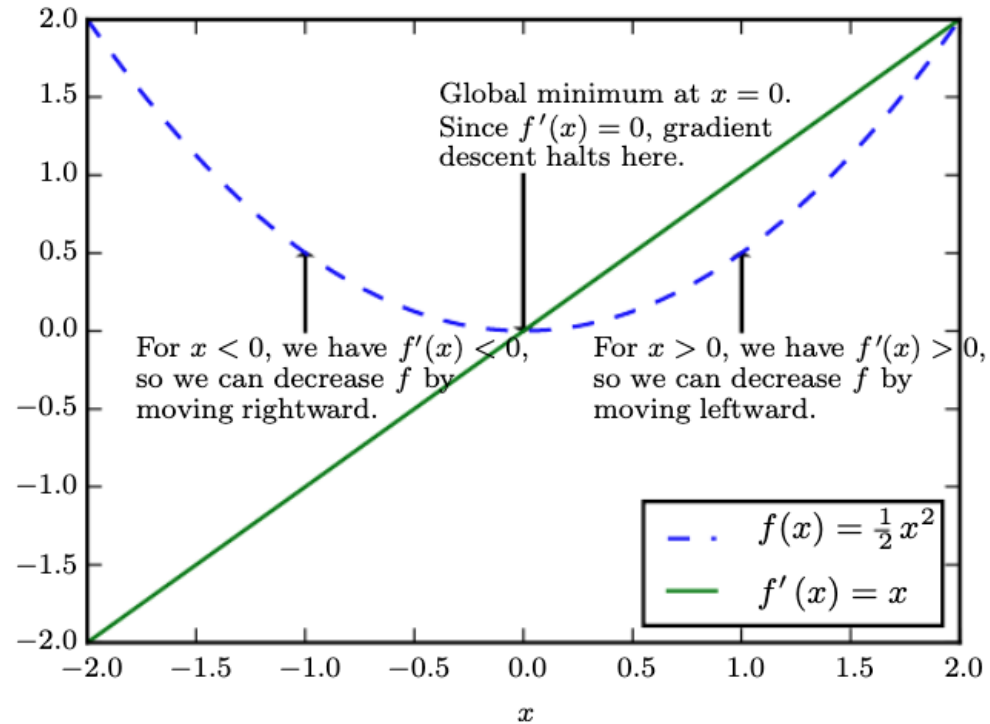


(b) After applying dropout.

Dropout: A Simple Way to Prevent Neural Networks from Overfitting

Optimization

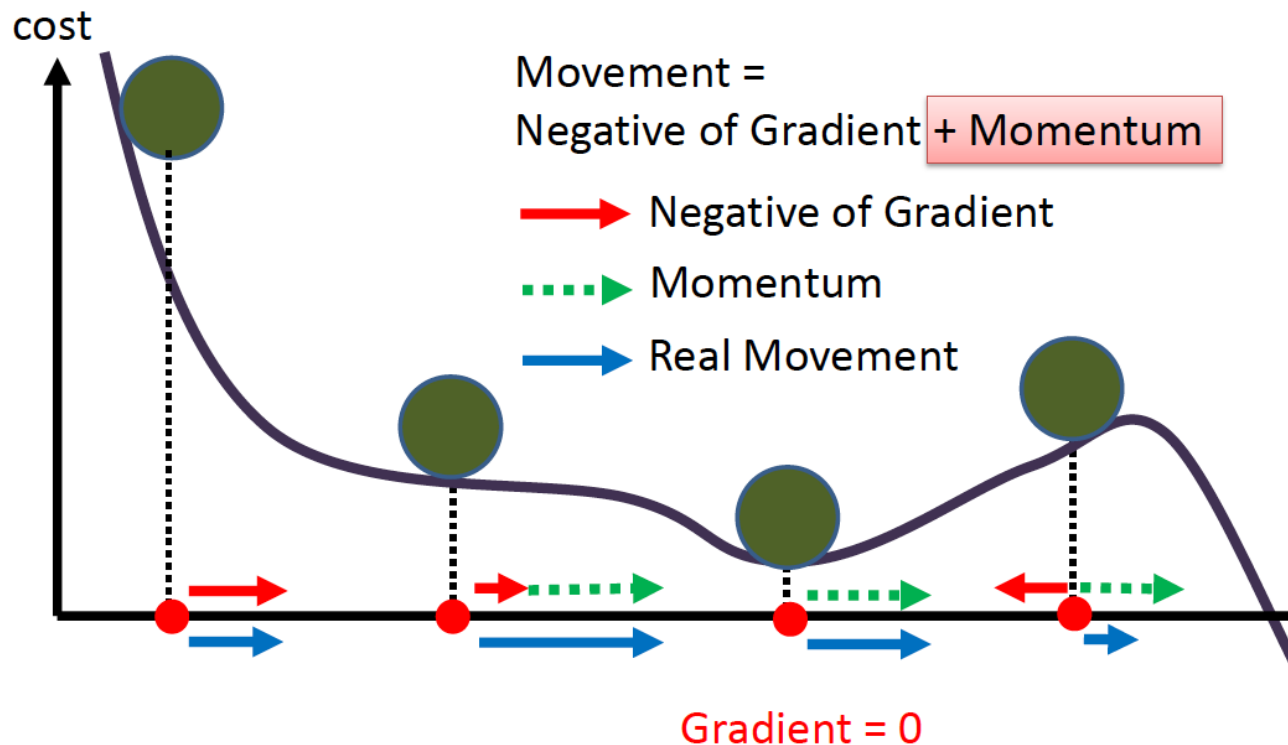
- Method to minimize the **cost function** by updating weights
- Gradient descent:
 - Iteratively moving in the direction of steepest descent as defined by the negative of the gradient
- Stochastic gradient descent (SGD)
 - To handle large training sets
 - Only run a subset of the training sets (i.e., batch/minibatch) for each update
 - Easier to converge



Optimization

■ Momentum:

- Prefers to go in a similar direction as before



SGD

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

```
while True:
    dx = compute_gradient(x)
    x -= learning_rate * dx
```

SGD+Momentum

$$v_{t+1} = \rho v_t + \nabla f(x_t)$$

$$x_{t+1} = x_t - \alpha v_{t+1}$$

```
vx = 0
while True:
    dx = compute_gradient(x)
    vx = rho * vx + dx
    x -= learning_rate * vx
```

Further Reading

The AlexNet Paper

ImageNet Classification with Deep Convolutional Neural Networks

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